ANALYSTLAB AFRICA

Experience Lab Internship Programme | Week 3

Hospital Operations, Cost & Patient Flow Intelligence

Track A: Data Analytics | Healthcare Operations & Financial Intelligence Dashboard

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Task 1: Understanding the Business Problem

Background

The healthcare organization wants to understand how its hospital operates: how many patients it sees, what kinds of visits they make, what those visits cost, how much of the cost payers cover, and which procedures are performed most. The data provided covers encounters, patients, procedures, payers and one organization, from January 2011 to February 2022.

1. The problem to investigate

The organization holds detailed records but has no single view of how encounter activity, claim costs, payer coverage and procedures relate to one another. This work answers the following questions:

- How have encounter volumes changed over time?
- Which encounter classes account for most encounters, and which cost the most?
- How much of the total claim cost is covered by payers, and how many encounters have no recorded coverage?
- Which procedures occur most often, and which have the highest average base cost?

2. Why it matters

Encounter volume and class mix show where hospital capacity and staffing are being used. Claim costs and payer coverage show where the hospital is financially exposed. In this data a large share of encounters has zero recorded payer coverage, so understanding who pays, and how much, directly affects financial planning. Procedure frequency and cost show where resources are concentrated.

3. Intended user

Hospital operations managers and finance and planning leads, who need a clear and trustworthy view without working through raw records. They are not data specialists, so the dashboard must be easy to read and its metrics clearly defined.

4. Decisions the findings could support

- Planning staffing and capacity around volume trends and the busiest encounter classes.
- Reviewing the payer mix and the encounters with no recorded coverage.
- Prioritizing cost review for the encounter classes and procedures that drive the highest costs.

5. Limitations of the available evidence

- The data covers a single hospital, so findings cannot be generalized to other organizations.

- 2022 is a partial year (data ends 5 February 2022), which can distort trend comparisons. 2011 has the lowest volume of the full years, and the data does not explain why.
- Zero payer coverage may mean uninsured, unrecorded or unpaid. The data does not say which, so it is reported without assuming a cause.
- Only about half of encounters have a linked procedure record, so procedure analysis covers part of the activity.
- Total claim cost and base cost measure different things, and the data does not fully explain how claim cost is built up.
- Only 75 encounters lasted over 24 hours, and the measure is affected by extreme values: the median encounter lasts 15 minutes, and 13 of the 75 last over 30 days. It therefore mixes long stays with long-running records, and conclusions about long stays rest on a small number of cases.
- Procedure start and stop times fall outside the time window of their encounter in about 51% of cases, so procedures are placed in time using their encounter's date, not their own timestamps.
- The records appear to be synthetic or anonymized, so findings show how the method works rather than real-world performance.
- The data contains no clinical outcomes, so no clinical claims or treatment recommendations are made.

Task 2: Dataset Inspection

A. Tables and fields used

Table	Fields used	Purpose
encounters	Id, START, STOP, PATIENT, PAYER, ENCOUNTERCLASS, BASE_ENCOUNTER_COST, TOTAL_CLAIM_COST, PAYER_COVERAGE	Main fact table: volumes, classes, costs, coverage
procedures	ENCOUNTER, PATIENT, CODE, DESCRIPTION, BASE_COST	Procedure frequency and cost
payers	Id, NAME	Payer analysis
patients	Id, GENDER, BIRTHDATE (age only)	Optional demographics; identifying fields removed
organizations	Id, NAME	Context only (one organization)

B. Records and columns

File	Records	Columns
encounters.csv	27,891	14
patients.csv	974	20
procedures.csv	47,701	9
payers.csv	10	7

File	Records	Columns
organizations.csv	1	8
data_dictionary.csv	65	3

C. Missing values, duplicates, data types and dates

Issue	Finding	Decision
Missing values: encounters	REASONCODE and REASONDESCRIPTION are blank in 19,541 rows (70%).	Left as is. Blank means no recorded reason; these fields are not used.
Missing values: procedures	The same two fields are blank in 36,945 rows (77%).	Left as is.
Missing values: patients	DEATHDATE blank for 820, ZIP for 142, SUFFIX for 953, MAIDEN for 588, MARITAL for 1. 154 patients have a death date.	Fields not used in the analysis; no action.
Missing values: payers	One payer (NO_INSURANCE) has no address, city, state, zip or phone.	Expected, as it is not a real insurer.
Duplicates	None: no duplicate rows, no duplicate encounter Ids, no duplicate procedure rows.	Nothing removed.
Data types	START and STOP are text in UTC format (e.g. 2011-01-02T09:26:36Z). ZIP is stored as a decimal in patients and payers.	Convert START and STOP to date/time in Power Query. Treat ZIP as text if used.
Date issues	No stop-before-start timestamps and no blank times. Range is 2 Jan 2011 to 5 Feb 2022.	Create a date table. Flag 2022 as incomplete.
Duration outliers	Median encounter duration is 15 minutes. 75 encounters exceed 24 hours; 13 of them exceed 30 days and 4 exceed one year (maximum 44,930 hours, about 5.1 years).	Kept. The over-24-hours measure is reported with this caveat.
Procedure timestamps	In 24,388 procedures (51.1%) the start time falls outside the time window of their encounter (2,060 before it starts, 22,328 after it ends).	Procedure timestamps are not used for timing. Procedures are dated by their encounter's start date.
Costs	89 encounters have a total claim cost of zero. Payer coverage never exceeds the claim cost.	Kept; noted in the data-quality assessment.

D. Relationships between tables

- encounters[PATIENT] to patients[Id] (many to one): 0 orphan records.
- encounters[PAYER] to payers[Id] (many to one): 0 orphan records.
- encounters[ORGANIZATION] to organizations[Id] (many to one).
- procedures[ENCOUNTER] to encounters[Id] (many to one): 0 orphan records.

- Every procedure's patient matches the patient on its encounter, and every patient has at least one encounter.
- 13,221 encounters have no procedures. This is valid but limits procedure analysis.

E. Limitations that could affect validity

- **Zero coverage:** 13,586 encounters (48.7%) have no recorded payer coverage. 8,807 of them are with the NO_INSURANCE payer (64.8% of the zero-coverage encounters), and every NO_INSURANCE encounter has zero coverage. The other 4,779 are with named insurers. The data does not explain why, so no cause is assumed.
- **Totals:** Total claim cost is about $101.5M and total payer coverage about $31.1M (30.6%).
- **Trend caveats:** Encounters per year range from 1,336 in 2011 (the lowest full year) to a peak of 3,885 in 2014, and only 220 in 2022 because the data ends on 5 February 2022. The 2022 figure should not be read as a real decline, and differences between years should be read with care, since the data does not explain what drives them.
- **Data dictionary:** It matches the files except that Middle and FIPS County Code are listed for patients but are not in patients.csv.
- **Single organization:** All data comes from one hospital.
- **Other limits:** Only 75 encounters lasted over 24 hours and some of them are extreme (13 exceed 30 days), procedure records cover only part of the encounters, procedure timestamps do not align with encounter times, and the components of claim cost are unclear.

F. Responsible data handling

patients.csv contains names, addresses and birthplaces. These columns will be removed in Power Query and will not appear in the dashboard, screenshots or any shared file. No clinical claims or treatment recommendations will be made. The provided SQL setup script drops and recreates a database named hospital_db, so it will only be run in an isolated practice environment after inspection.

Task 3: Success Criteria

The following measurable criteria will be used to judge whether the work is successful. Each can be checked with Power BI, Excel or SQL.

#	Criterion	Measure	Target	How it is verified
1	Dashboard totals reconcile with an independent check	Total encounters, total claim cost and total payer coverage in Power BI compared with Excel and SQL	Zero difference (27,891 encounters; $101,514,375.52 claim cost; $31,097,506.99 payer coverage)	Excel pivot tables and SQL COUNT and SUM queries on the same data
2	Totals respond correctly to filters	Sum of encounters and claim cost across all values of a filter (year, encounter class, payer) compared with the unfiltered total	100% match for every filter tested	Apply each slicer in turn and record the results in a validation table

#	Criterion	Measure	Target	How it is verified
3	Data model integrity	Orphan records on each relationship, and duplicate keys	0 orphans and 0 duplicate keys on all relationships	Power Query checks and the relationship view in Power BI
4	KPI definitions are reproducible	Number of KPIs with a written definition (field, formula, filters, DAX)	At least 5 of 5 documented; another analyst can recalculate them	KPI definition sheet, with one KPI recalculated by hand from its definition
5	Required analysis is covered	Number of the eight assignment questions answered with a metric, calculation method and interpretation	At least 4 (5 planned)	Checklist against the assignment question list
6	Dashboard completeness	Required pages present: overview, volume trend, encounter-class comparison, cost and payer coverage, procedure utilization, with filters	5 of 5 pages	Review against the assignment checklist
7	Privacy and responsible handling	Personal identifiers visible in the dashboard, screenshots or shared files	0	Review of every page and screenshot before submission

Note: the reconciliation figures in criterion 1 come from an initial check of the source CSV files and will be confirmed independently in Excel and SQL during validation (Task 4).

Task 4: Professional Standards

1. Project folder and file names

```
AnalystLab_Week3_Hospital_Dashboard/
  01_Raw_Data/            original CSV and SQL files, never edited
  02_Working_Data/        cleaned or exported tables, if any
  03_PowerBI/             Hospital_Operations_Dashboard_v1.pbix
  04_SQL/                 validation_queries.sql
  05_Excel_Validation/    Validation_Checks.xlsx
  06_Documentation/       Tasks_1_to_4.docx, Data_Quality_Assessment.docx,
                          KPI_Definitions.xlsx, Assumptions_Log.docx
  07_Screenshots/         01_Overview.png, 02_Trend.png, ...
  08_Final_Submission/    Executive_Summary.docx and final files
```

File names use no spaces, a clear subject and a version number (v1, v2), and are used consistently wherever files are referred to.

2. Assumptions and decisions log

#	Assumption or decision	Reason
1	Zero payer coverage is reported as recorded, with no cause assumed.	The data does not say whether it means uninsured, unrecorded or unpaid.
2	No duplicates or missing values were removed or replaced.	No duplicate records exist; the blank fields are not used in the analysis.
3	Encounter duration is calculated as STOP minus START in hours.	Both timestamps are valid and in the same time zone (UTC).
4	2022 is flagged as a partial year in trend visuals, and year-on-year change is not shown for it.	The data ends on 5 February 2022.
5	Total claim cost is used for cost analysis; base cost is kept separate.	They measure different things.
6	Procedure analysis covers only encounters with a procedure record.	13,221 encounters have no procedure.
7	Personal columns are removed from patients.	Privacy.
8	Procedures are placed in time using their encounter's start date, and procedure start and stop times are not used.	In 51% of procedures the start time falls outside the encounter's time window.
9	The over-24-hours KPI is reported with a caveat and no extreme values are removed.	13 of the 75 encounters exceed 30 days; the median encounter lasts 15 minutes.

Further decisions made in Power Query and DAX will be added to this log with the reason for each.

3. Source attribution

- Data: the healthcare dataset pack supplied by AnalystLab Africa (encounters, patients, procedures, payers, organizations, data dictionary).
- Tools: Power BI, Power Query, DAX, Excel and SQL.
- Any code, DAX pattern or guidance adapted from a tutorial, book or AI tool is noted here with its source. AI assistance (Claude, by Anthropic) was used for planning, data profiling and drafting documentation.

4. Evidence of testing and validation

A validation table will record the date, the check, the Power BI result, the independent result (Excel or SQL), the difference and the status. Screenshots of the Excel and SQL results will be saved beside the matching Power BI figures in the Excel validation folder.

5. Privacy

- Names, addresses and similar identifiers are removed in Power Query before loading.
- Every page, tooltip, screenshot and file is reviewed before submission for patient names, addresses or IDs.
- Patient IDs are not displayed; only counts are shown.
- Raw data and the .pbix file are not uploaded to any public repository.

6. No unsupported clinical claims

Findings are stated as operational and financial observations only. No statements are made about diagnosis, treatment quality or the care patients should receive.

7. Database safety

The provided create_hospital_db.sql script drops and recreates a database named hospital_db. It is inspected before use and run only in an isolated practice environment, never against an existing database. Whether it was run is recorded in the assumptions log.